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Project Title:	Cell:Cell Mediated Assembly of Cardiac Tissue in Microgravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19899	**Congressional District:**	1
Comments:			
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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	This project addresses aspects of cardiac tissue engineering. The capacity of isolated rat and human heart cells to reconstitute cardiac tissue in culture without extensive cues from three-dimensional scaffolds or matrices is being investigated. The approach takes full advantage of the simulated-microgravity environment provided by NASA-bioreactors to allow isolated cells to re-establish tissue structure in vitro. The proposed research also investigates the role of gravity and the bioreactor environment in the structural and molecular/biochemical phenotype of cardiac cell assemblages. A combination of morphological, cell biological, molecular, and biochemical methods will be used to characterize cardiac cell cultures and to compare bioreactor-derived cultures to native tissue and to cultures grown using standard tissue culture methods. In the long-term, it is hoped that this study will lead to the ability to engineer biosynthetic cardiac patches for the treatment of a wide variety of cardiac abnormalities. Further, it is hoped that this research will lead to an increased understanding of the influence of gravity on cells and cell assemblages in culture. The work investigates the ability of isolated heart cells to re-establish structural and molecular/biochemical phenotypes of		

Task Description:	heart tissue when cultured in microgravity environments. In particular, the study will: 1)Evaluate the process by which cardiac-specific structures are formed by isolated rat and human cardiac cells. 2)Identity the extra-cellular matrix (ECM) material that is differentially produced in bioreactor-based cultures. 3) Assess the role of gravity in determining the distribution of subcellular organelles in cultured heart cells. 4) Evaluate the effect of bioreactor-based culture on signal transduction pathways. 5)Determine the changes in gene expression caused by culture in microgravity using differential display RT-PCR. For Spaceflight: The long-term presence of humans in space will require that critical medical and surgical procedures be conducted at remote sites utilizing limited resources. Of particular concern will be the ability to maintain adequate circulatory function, which is required to sustain life and to allow crew-members to remain productive during flight and thereafter. Among the issues of relevance to NASA's spaceflight programs, two are critical. First, physical and pharmacologic countermeasures that allow crew members to remain productive during shifting hemodynamic-loading conditions associated with changes in gravity are needed; strategies that require limited additional "upweight" would be especially useful. Our studies focus on pathways that impact the organization and function of cardiac muscle in order to develop pharmacologic interventions that will improve circulatory function under changing conditions or in disease. We have identified a number of potentialyy importnat pathways including a heme-metabolic pathway that may be amenable to pharmacologic manipulations using simple and naturally-occurring compounds. Stimulation of this pathway improves cardiac muscle cell organization and function. Second, there is a need to develop strategies that reverse or repair profound heart disease that may arise in crew members. Heart disease, which is a leading cause of morbidity and mortality in humans, may occur as a direct result of the spaceflight environment, as a secondary result of accommodation to long-term mission conditions, or as a consequence of unanticipated disease, injury, or infection. Our studies are directed toward the development of implantable constructs produced on site using defined media with primary isolated cells and simple bioreactors. Thus, although at an early stage, our research may be highly relevant to the development of strategies supporting the long-term presence of humans in space. On Earth: Congenital and acquired heart diseases are significant problems. The treatments that are currently available mainly address the sequelae of primary defects and often fall short of full structural and functional recovery. In addition, there is a severe shortage of donor tissue for reconstructive cardiac surgery and transplantation. Advances in treatment are hampered by a lack of suitable material and a lack of characterization of the molecular pathways that establish and sustain tissue structure. New treatments for heart diseases, new sources of implant materials, and accessible models for the study of cardiac morphogenetic pathways are all needed, and tissue-engineering has emerged as a potential means to address these areas. Cardiac tissue engineering is an emerging field involving the construction of tissue equivalents in vitro. Using an approach in which self-assembly is encouraged, work by the applicant shows that isolated neonatal rat heart cells (NRHCs) re-establish characteristics of three-dimensional cardiac tissue when grown in specialized bioreactors. Through the use of NASA biotechnologies, our knowledge of the mechanisms and pathways involved with in vitro tissuegenesis is being extended so that an accessible source of tissue-equivalent material can be developed. The development of tissue-equivalents for use as research models or for use as implant devices could profoundly alter the course of cardiac-related research and dramatically improve the outlook for patients with a variety of heart diseases.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Our lab studies the formation of cardiac tissue structures in culture using primary heart cells and rotating-wall bioreactors. Isolated neonatal rat heart cells (NRHCs) grown in bioreactors re-establish 3D architecture that is similar to native tissue. This tissuegenesis occurs in the absence of 3D organizational cues from biomaterials and involves the migration of endothelial cells (ECs) from their initial culture positions. Since Notch and BMP2 signaling are involved in EC migration during heart development, during this past year we tested the hypothesis that the self-assembly of cardiac tissue architecture in vitro involves up-regulation of Notch and BMP2 pathways. The expression of Notch/BMP2 and their pathway intermediates during the formation of cardiotypic structures was studied using real-time PCR and Western Blotting and evaluated by comparison of bioreactor expression profiles with those seen in standard plate culture. Real-time PCR analyses of Notch receptor (Notch1, Notch2, and Notch3), Notch ligand (Jagged1), and Notch downstream target (Hes1), as well as BMP2, all indicated a pronounced increase in mRNA level during the initial days of growth in bioreactor culture; the level of each gene product remained constant in plate-grown cells. Western blotting verified the relative increase in Jagged1 protein during early bioreactor culture and confirmed the increase in Notch1 signaling activity within the same time period. Efforts are underway to elucidate the specific function of the Notch and BMP2 pathways in bioreactors by studying the effects of their disruption on structure formation in bioreactor cultures and on EC migration within bioreactor-derived constructs. Our results indicate that Notch signaling and BMP2 expression are elevated early in the in vitro cardiac tissuegenesis process. Manipulation of these pathways may allow the formation of functional tissue architecture to be controlled in the production of cardiac tissue equivalents for clinical and research applications. In addition, we have found that induction of heme oxygenase-1 (HO1 or HSP-32) activity improves cardiomyocyte organization and function in vitro. We are continuing to elucidate the specific mechanism(s) by which this occurs and have found that CO production by HO in cardiac cells may be required to mediate the positive effects of HO. Accessing this pathway in vitro may allow tissue engineers to control the growth and function of cardiac tissue constructs.
Bibliography Type:	Description: (Last Updated: 06/28/2012)
Articles in Peer-reviewed Journals	Akins R Jr, McLaughlin T, Boyce R, Gilmour L, Gratton K. "Exogenous metalloporphyrins alter the organization and function of cultured neonatal rat heart cells via modulation of heme oxygenase activity." J Cell Physiol. 2004 Oct;201(1):26-34. PMID: 15281086 , Oct-2004
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Presentation	Akins, R., McLaughlin, T., Boyce, R., Gilmour, L., and Gratton, K. "Heme Oxygenase Improves the Organization and Function of Cultured Neonatal Rat Heart Cells" NASA Cell Science Conference Feb-2004
Presentation	Bruner, M. and Akins, R. "Role of Heart Developmental Genes in Assembly of 3D Cardiac Tissue Structures In Vitro" NASA Cell Science Conference. Feb-2004
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Presentation	Akins, R.E. "Controlling Cell Phenotype and Multi-Cellular Organization in Cardiac Tissue Engineering and Regenerative Therapies" Department of Biological Sciences, University of Delaware Nov-2004